

DHRUBA KHANAL

SQL/ETL DEVELOPER

682-477-8832

khanalDK2042@gmail.com

Fort Worth, TX, 76131

SUMMARY

- Engrossing grand 5 Years of experience with passion for building end to end direct capabilities, writing scalable code, as well as debugging integration packages, pipelines required for optimal ETL of data from a wide variety of sources.
- Extensively worked on MS SQL Server 2022/2019/2017/2016 MSBI Technologies (SSIS, SSAS and SSRS), C# and .NET. Expertise in various phases of project life cycles (Design, Analysing, Implementation and testing).
- Highly experienced in software development life cycle, business requirement analysis, design, programming, database design, data warehousing and business intelligence concepts, Star Schema and Snowflake Schema methodologies.
- Experienced on PaaS Services for Data ingestion, transformation, and storage of data from OnPrem to Cloud and vice versa, worked on Cost Optimization by porting live reporting data to OnPrem from Cloud storage partitioned based on time series.
- Conversant with Data Distribution algorithms like Round Robin, ASCII and Replicator, implemented Data Partitioning using Round Robin in Data Warehouse built using Azure MS SQL PaaS service.
- Well experienced on Server less Azure Data Factory Author and Monitor features in order to build Cost efficient pipelines to port data from multiple sources.
- Implemented Azure Data Factory Integration Runtimes to execute SSIS packages, collecting OnPrem data scenarios, Parametrized Linked Services and re used existing Data Sets across most of the activities in a Pipeline.
- Experience with logical and physical data modelling and database design, handling granularity, cardinality to optimum performance and storage.
- Well versed in Data Modelling, Design, Development, implementation & Administration of Database systems using MSSQL 2022/2019/2017/2016 for both OLTP (Batch Processing, Online Processing) & Data warehousing Systems (SSRS, SSIS, SSAS).
- Proficient in programming complex queries using stored procedures, Triggers, Cursors, User Defined Functions using T – SQL to handle complex business scenario.
- Partitioning tables at different custom ranges to improve performance of data loading in tables using SQL Server Enterprise edition.
- Highly experienced in Dimension modelling using star and snowflake schemas, handling slowly changing dimension using different techniques like row hash (MD5 and SHA), slowly changing dimension wizard and Lookup, especially in incremental load of data.
- Experienced in handling late arriving dimension and late arriving facts scenarios in a Data Warehouse.
- Experienced in Designing, Creating and processing of cubes using SSAS. Created and Configured Data Sources and Data Source Views, Cubes, and calculated measures for OLAP cubes.
- Worked on defining Measures Groups and Dimensions by analyzing Data Warehouse in order develop multidimensional cubes and tabular models in SSAS and Responsible for developed MDX queries for faster retrieval historical data from cubes.
- Developed and Configured Data Sources and Data Source Views, Dimensions, Hierarchies – Levels, Measure Groups, Dimension Usage Tab to define relation between Dimension and Measure Groups, Calculated Measures, KPIs and Storage Partitions in OLAP Cubes.
- Created different Power BI reports utilizing the desktop and the online service and schedule refresh.

TECHNICAL SKILLS

RDBMS Servers: MS SQL Server 2022/2019/2017/2016

Cloud Utilities: Azure Data Factory Studio, Azure Portal, Azure IAM, Azure CI /CD platform, Azure MS SQL and Apache SuperSet.

Database Utilities: SQL Server Management Studio, SQL Profiler, Query Analyzer, Visual Studio

ETL Utilities: SQL Server Integration Service (SSIS)

Data Analytics Utilities: SQL Server Analysis Service (SSAS)

Reporting Utilities: SQL Server Reporting Service (SSRS) Microsoft Power BI, MS-Excel

Programming Utilities: T-SQL, C#, C++, VB.net

Web Technologies: HTML, Java Script

Operating Systems: Windows Server DOS, Linux

Other Utilities: MS Office Suite (Microsoft Word, Power Point, Microsoft Visio), ERWIN.

PROFESSIONAL EXPERIENCE

SQL/ETL Developer (December 2024 – PRESENT)

Educology Solutions, Washington, DC

Responsibilities:

- Re-engineered legacy data warehouse evolved over time to migrate data warehouse data model on Azure Synapse Analytics using redesign strategy.
- Reduced the complexity of existing legacy data warehouse before migrating, been cognizant of all ways to simplify existing data warehouse and document them.
- Removed and archived unused tables before migrating, avoided data which is no longer in use, helped to reduce the overall data volume to migrate.
- Converted physical data marts to virtual data marts, minimized what to migrate to reduce the total cost of ownership, Optimized redundant dimension, measures into unified entities by remodeling schemas.
- Re-designed existing SSIS ETL packages in the Azure Data Factory Pipelines to load data from multiple sources and staging sinks, Populating data from staging to Azure Synapse.
- Programmed API calls to Azure Data Factory APIs to orchestrate nightly loads of Data Warehouse and Data Marts.
- Prepared checklist to help collect necessary information to validate migration – Row counts, tables and indexes, Data compression ratios, Partitions, Types and Code objects (Functions, Stored procedures).
- Working on Postman, Gateways, JWT Tokens, OAuth Protocols as part of debugging API request payload and response codes, Registering Clients in Azure Entra ID (Active Directory) and Fetching Client Secrets from Azure Key Vault.
- Explored Azure ExpressRoute to make private connection between on prem sources and Azure Data centres, AzCopy, Azure DataBox and SqlBulkCopy API to copy files and blobs to/from storage accounts.
- Handled common migration issues and resolutions, Unsupported data types and workarounds, Integrity constrain differences, SQL incompatibilities and resolutions.
- Implemented Dynamic Data Masking (DDM) used to obfuscate data, Encryption using TLS and SSH.

- Been part of DevOps team to guide deployment strategies and plans using ARM template, Integration Run-times and Python scripts.

Technical Environment: Microsoft SQL Server 2022 Enterprise Edition, Azure Synapse, Azure Data Factory V2, SQL Server Management Studio (SSMS) and GIT.

SQL DEVELOPER (MAR 2022 to Nov 2024)

FEDEX SUPPLY CHAIN - COPPELL, TX

Responsibilities:

- Liaised with business stakeholders and product teams to gather, analyze, and translate functional and technical requirements into comprehensive design and specification documents aligned with strategic objectives and delivery timelines.
- Designed and implemented complex relational database architectures, developing tables, views, stored procedures, user-defined functions, indexes, and triggers in T-SQL, ensuring optimal performance and adherence to normalization standards.
- Enhanced and refactored legacy database components through detailed code reviews, query optimization, and indexing strategies, improving scalability, performance, and maintainability across critical data systems.
- Developed advanced T-SQL queries and SQL scripts to enable complex business logic, high-volume data transformations, and seamless application integration across enterprise platforms.
- Architected and engineered robust ETL solutions using SSIS, incorporating data flow, control flow, exception handling, and auditing frameworks through Execute SQL, Data Flow, Bulk Insert, FTP, Send Mail, and Loop containers.
- Devised and implemented end-to-end ETL strategies for integrating data from SQL Server, Oracle, Sybase Adaptive Server, and flat files, ensuring data quality, consistency, and accuracy during migrations and legacy system transitions.
- Automated ETL orchestration using .NET scripting and configuration-driven workflows, optimizing performance, parameterization, and reusability across diverse data integration pipelines.
- Deployed and scheduled SSIS packages via SQL Server Agent, managing production workloads, incremental updates, and performance monitoring to maintain high data availability and process reliability.
- Developed and deployed interactive BI reports using SSRS, including Matrix, Tabular, Drill-down, Drill-through, Cascading, and Parameterized reports, leveraging advanced expressions and dynamic filtering for actionable insights.
- Documented and validated SSRS report logic and design, reviewing peer-developed reports to ensure accuracy, efficiency, and compliance with data presentation standards and business requirements.
- Designed and developed OLAP cubes using SSAS, creating dimensions, hierarchies, measures, and aggregations with star and snowflake schema models to support multi-dimensional analysis and executive dashboards.
- Led the design and integration of enterprise Business Intelligence solutions using Informatica, Business Objects, Cognos, and Erwin Data Modeler, standardizing metadata, data lineage, and analytical reporting frameworks.
- Conducted comprehensive manual and Agile-based testing, including unit, integration, and data validation tests, utilizing WinRunner and managing defect tracking and closure through Test Director.

- Maintained end-to-end project documentation and version control, ensuring deployment consistency, traceability, and adherence to governance standards across all environments.
- Delivered advanced business insights through Power Pivot and SharePoint integration, enabling collaborative analytics, report scheduling, and enterprise data accessibility.

Technical Environment: SQL Server 2019, T-SQL, Visual Studio, SQL Server Integration Services (SSIS), SQL Server Reporting Services (SSRS), Windows Server 2019, SQL Server Management Studio (SSMS), Visual Source Safe, EXCEL, AGILE, SQL Server Profiler, .Net, MS Word, Visio, Oracle, Database engine Tuning advisor, developer

SQL/ETL DEVELOPER (JULY 2021 to FEB 2022)

TEXAS HEALTH FAMILY CARE - HIGHLAND VILLAGE, TX

Responsibilities:

- Interacted with business analysts and data architects to interpret functional requirements and transform them into well-structured database and ETL design specifications supporting analytical and transactional workloads.
- Developed database components including stored procedures, functions, triggers, views, cursors, and indexes in T-SQL to support complex business logic and application integration requirements.
- Optimized database performance through index tuning, query optimization, and execution plan analysis, leveraging index seek/scan analysis and Database Engine Tuning Advisor to enhance overall query efficiency.
- Designed and developed SSIS packages to extract, transform, and load (ETL) data from heterogeneous sources such as SQL Server, Sybase Adaptive Server, flat files, and Excel into staging and data mart layers.
- Created and maintained data flow and relational models in Microsoft Visio, aligning database structures with data warehouse architecture and business reporting objectives.
- Implemented complex transformations in SSIS, including Lookups, Conditional Splits, Derived Columns, and Data Conversions, performing extensive data cleansing and validation before loading to the staging or target systems.
- Automated SQL Server maintenance operations by building SSIS packages for index rebuilds, database backups, and statistics updates, integrating error handling and logging mechanisms via SSIS Event Handlers.
- Participated in end-to-end data mart development, supporting both OLTP and OLAP environments through incremental ETL design, dimensional modeling, and metadata management.
- Developed and deployed SSRS reports to deliver actionable insights in multiple formats (Word, Excel, PDF) creating RDLs and parameterized reports for account aggregation and financial data visualization.
- Designed and published ad-hoc reports using Report Builder, empowering business users to generate customized reports through preconfigured relational data templates.
- Developed OLAP cubes in SSAS, identifying fact-dimension relationships, defining KPIs, partitions, and perspectives, and implementing drill-through and calculated measures to enhance analytical usability.
- Created and optimized Power BI dashboards and reports, leveraging DAX, Power Query, Pipelines, and Paginated Reports, and utilized DAX Studio for performance tuning and complex measure development.
- Performed database and report validation testing, developing SQL-based test cases to verify ETL logic, data quality, and performance across staging and reporting layers.
- Contributed to BI solution development using Informatica and SSIS, ensuring seamless data integration, incremental updates, and consistency across enterprise data platforms.

- Documented ETL workflows, process runtimes, and test results, ensuring traceability, repeatability, and compliance with internal development and deployment standards.

Technical Environment: SQL Server 2019, SSIS 2019, SSRS 2019, SSAS 2019, Power BI (2021), Power Query, DAX Studio, Sybase Adaptive Server 15.7, Informatica 10.4, Erwin Data Modeler 2022, Microsoft Visio 2021, Report Builder 3.0, Visual Studio 2019, Windows Server 2019.

JR SQL DEVELOPER (FEB 2020 to MAY 2021)

EECU CREDIT UNION - HURST, TX

Responsibilities:

- Assisted in gathering business and technical requirements to understand data flow processes, reporting objectives, and integration needs across multiple business domains.
- Installed, configured, and supported Microsoft SQL Server 2014 and Visual Studio 2015 on network domain environments to establish the development and testing infrastructure for database and BI solutions.
- Participated in the design and development of logical and physical data models using Erwin Data Modeler 4.2, collaborating with senior developers and data architects to align schema structures with business rules and performance objectives.
- Analyzed data quality and consistency issues in coordination with the data warehouse architect, contributing to the creation of standardized data validation and profiling procedures.
- Developed and implemented SSIS packages to migrate, cleanse, and transform data between production, staging, and development environments, ensuring accuracy and referential integrity during transfers.
- Designed and configured ETL workflows using SSIS to extract data from diverse sources including SQL Server, Oracle, Excel, Text, and Access files, and load it into data marts for analysis and reporting.
- Enhanced ETL package performance through parameterization, configuration files, and logging mechanisms to support dynamic package deployment across multiple environments.
- Built and maintained a consolidated data store to centralize business information for analytical and reporting purposes, ensuring consistent and timely data refresh cycles.
- Developed SSRS reports including parameterized, drill-down, and aggregate reports, enabling users to perform trend and comparative analysis across operational and financial data sets.
- Defined report layouts, datasets, and parameters in SSRS, ensuring that generated daily, weekly, and monthly management reports met accuracy and visualization standards.
- Contributed to report deployment and scheduling processes, ensuring secure delivery of business intelligence reports via SSRS to various stakeholders.
- Performed basic data validation and functional testing on ETL processes and SSRS outputs to confirm data completeness, accuracy, and consistency across environments.
- Documented ETL workflows, report generation procedures, and deployment steps, maintaining project traceability and supporting knowledge transfer within the team.
- Collaborated with senior engineers during system analysis and code reviews, gaining hands-on exposure to query tuning, data modeling principles, and BI development best practices.

Technical Environment: SQL Server 2016, SSIS 2016, SSRS 2016, Visual Studio 2016, Erwin Data Modeler 4.2, Oracle 11g, Microsoft Access , Microsoft Excel , Windows Server 2016 and VSTS.